

Andrew Fowlie

Associate Professor | Physicist & Data Scientist

Researcher in statistical inference, computational methods and scientific software, with applications in physics. 60+ papers; 3000+ citations; 50+ presentations. Research contributions include Bayesian inference, simulation-based inference, model selection, uncertainty quantification and open-source software.

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Areas of specialization

- Bayesian inference, statistical modelling and computational methods, including parameter estimation, model selection and scientific software
- Data analysis and statistical inference for high-energy physics, including collider experiments, dark matter, supersymmetry and Higgs physics
- Model comparison, fine-tuning, naturalness and the hierarchy problem

Experience

- 2023 – Associate Professor (2025 –); Assistant Professor (2023 – 2024)
School of Maths and Physics, XJTLU, Suzhou, China
- 2018 – 2022 Associate Professor, Department of Physics, Nanjing Normal University, Nanjing, China
- 2015 – 2018 Postdoctoral researcher, Department of Physics, Monash University, Melbourne, Australia
Particle phenomenology and Bayesian statistics with [Prof. Csaba Balázs](#)
- 2014 – 2015 Postdoctoral researcher, [KBFI \(National Institute of Chemical Physics and Biophysics\)](#), Estonia
Particle phenomenology with [Prof. Martti Raidal](#)
- 2009 – 2013 Ph.D., University of Sheffield, UK
[Bayesian Approach to Investigating Supersymmetric Models](#) supervised by [Prof. Leszek Roszkowski](#)

Research

GRANTS

Approx. RMB 1.4M/€180K as PI and RMB 2.7M/€340K as CI, including NSFC and ARC awards.

- 740 000 RMB ■ **PI.** 560 000 RMB NSFC + 180 000 RMB host matching. NSFC RFIS-II (W2432006), *Percolation of first-order cosmological phase transitions in the early Universe*, 1 January 2025 – 31 December 2026
- 180 000 RMB ■ **PI.** Post-graduate Research Scholarship (PGRS FOSA2406017), *Bayes factor surface for searches for new physics*
- 100 000 RMB ■ **PI.** Research Development Fund (RDF-22-02-079), *Exact-Approximate methods for searches for new physics at the Large Hadron Collider*, 1 July 2023 – 30 June 2025
- 550 000 RMB ■ **CI.** NSFC General Program (12275134), 暗物质粒子及其相关的新物理唯象学研究, [Wu, L.](#) (PI), Fowlie, A., et al., 1 January 2023 – 31 December 2026
- 449 659 AUD ■ **CI.** Australian Research Council Discovery Project ([DP210101636](#)), *Electroweak phase transition: A cosmological window to new particle physics*, [Kobakhidze, A.](#) (Primary Chief Investigator), [Balázs, C.](#) (Chief In-

vestigator), [Ramsey-Musolf, M.J.](#) (Partner Investigator), Fowlie, A. (Partner Investigator), 13 December 2021 – 12 December 2024

350 000 RMB

- **PI.** NSFC RFIS-I (11950410509), *Discovering dark matter with Bayesian and frequentist statistics*, 1 January 2020 – 31 December 2021

PUBLICATIONS

- *h*-index 31; 3000+ citations; 50+ publications
- First or corresponding author in *Nature Comm.*, *Nature Reviews Methods Primers*, *Rept. Prog. Phys.*, and twice in *Phys. Rev. Lett.*; published in *Prog. Part. Nucl. Phys.*
- Two papers with 250+ citations; six papers with 100+ citations & fifteen papers with 50+ citations
- See [INSPIRE](#) for complete publication data

Journal bibliometrics (percentile, quartile and CiteScore) are from [Scopus](#). Corresponding-author publications are marked with a star; author lists in particle physics are alphabetical.

- 2026 ★ [1] Cheek, A., **FOWLIE, A.** & Herrera, G. Quality control for axions and ALPs (2026). [[arXiv:2606.14098](#)], **1 cite**
- ★ Q1 [2] **FOWLIE, A.** It's all in your head—naturalness arguments do not require aleatoric uncertainty. *Synthese* [87%; Q1; CS: 2.4] **207**, 207, DOI: [10.1007/s11229-026-05573-2](#) (2026). [[arXiv:2604.18656](#)]
- 2025 Q2 [3] Athron, P. *et al.* Global fits and the search for new physics: past, present and future. *Sci. Bull.* [69%; Q2; CS: 2.6] **71**, 3765–3786, DOI: [10.1016/j.scib.2026.06.028](#) (2026). [[arXiv:2512.14109](#)], **3 cites**
- [4] Adhya, S. P. *et al.* CEPC Technical Design Report – Reference Detector (2025). [[arXiv:2510.05260](#)], **58 cites**
- Q2 [5] Ai, X. *et al.* New physics search at the CEPC: a general perspective. *Chin. Phys. C* [79%; Q2; CS: 5.1] **49**, 123108, DOI: [10.1088/1674-1137/ae1194](#) (2025). [[arXiv:2505.24810](#)], **36 cites**
- ★ [6] **FOWLIE, A.** PolyStan: PolyChord nested sampling and Bayesian evidences for Stan models (2025). [[arXiv:2505.17620](#)]
- ★ Q1 [7] **FOWLIE, A.** stanhf: HistFactory models in the probabilistic programming language Stan. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] **85**, 923, DOI: [10.1140/epjc/s10052-025-14495-1](#) (2025). [[arXiv:2503.22188](#)], **3 cites**
- ★ Q1 [8] Chang, C., Farmer, B., **FOWLIE, A.** & Kvellestad, A. Exact inference from noisy simulations in collider physics. *Phys. Rev. D* [91%; Q1; CS: 8.3] **113**, 072013, DOI: [10.1103/2d39-gsjp](#) (2026). [[arXiv:2502.08157](#)], **1 cite**
- 2024 Q1 [9] Athron, P., Balazs, C., **FOWLIE, A.**, Morris, L., Searle, W., Xiao, Y. & Zhang, Y. PhaseTracer2: from the effective potential to gravitational waves. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] (2024). [[arXiv:2412.04881](#)], **33 cites**
- Q1 [10] Albert, J., Balazs, C., **FOWLIE, A.**, Handley, W., Hunt-Smith, N., de Austri, R. R. & White, M. A comparison of Bayesian sampling algorithms for high-dimensional particle physics and cosmology applications. *Comput. Phys. Commun.* [93%; Q1; CS: 12.1] **315**, 109756, DOI: [10.1016/j.cpc.2025.109756](#) (2025). [[arXiv:2409.18464](#)], **7 cites**
- ★ Q1 [11] **FOWLIE, A.** & Herrera, G. Precise interpretations of traditional fine-tuning measures. *Phys. Rev. D* [91%; Q1; CS: 8.3] **111**, 015020, DOI: [10.1103/PhysRevD.111.015020](#) (2025). [[arXiv:2406.03533](#)], **6 cites**
- ★ Q1 [12] **FOWLIE, A.** The Bayes factor surface for searches for new physics. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] **84**,

- 426, DOI: [10.1140/epjc/s10052-024-12792-9](https://doi.org/10.1140/epjc/s10052-024-12792-9) (2024). [arXiv:2401.11710], **16 cites**
- 2023 **Q3** [13] Abdallah, W. *et al.* CEPC Technical Design Report: Accelerator. *Radiat. Detect. Technol. Methods* [42%; Q3; CS: 1.5] **8**, 1–1105, DOI: [10.1007/s41605-024-00463-y](https://doi.org/10.1007/s41605-024-00463-y) (2024). [arXiv:2312.14363], **277 cites TOPCITE100+**
- ★ **Q1** [14] Athron, P., **FOWLIE, A.**, Lu, C.-T., Morris, L., Wu, L., Wu, Y. & Xu, Z. Can Supercooled Phase Transitions Explain the Gravitational Wave Background Observed by Pulsar Timing Arrays? *Phys. Rev. Lett.* [94%; Q1; CS: 16.5] **132**, 221001, DOI: [10.1103/PhysRevLett.132.221001](https://doi.org/10.1103/PhysRevLett.132.221001) (2024). [arXiv:2306.17239], **108 cites TOPCITE100+**
- ★ **Q1** [15] **FOWLIE, A.** & Li, Q. Modeling the R -ratio and hadronic contributions to $g - 2$ with a Treed Gaussian process. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] **83**, 943, DOI: [10.1140/epjc/s10052-023-12110-9](https://doi.org/10.1140/epjc/s10052-023-12110-9) (2023). [arXiv:2306.17385], **5 cites**
- Q1** [16] Athron, P., Balázs, C., **FOWLIE, A.**, Morris, L. & Wu, L. Cosmological phase transitions: From perturbative particle physics to gravitational waves. *Prog. Part. Nucl. Phys.* [99%; Q1; CS: 24.5] **135**, 104094, DOI: [10.1016/j.pnpnp.2023.104094](https://doi.org/10.1016/j.pnpnp.2023.104094) (2024). [arXiv:2305.02357], **266 cites TOPCITE250+**
- Q1** [17] Ananyev, V. *et al.* Collider constraints on electroweakinos in the presence of a light gravitino. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] **83**, 493, DOI: [10.1140/epjc/s10052-023-11574-z](https://doi.org/10.1140/epjc/s10052-023-11574-z) (2023). [arXiv:2303.09082], **30 cites**
- ★ **Q1** [18] **FOWLIE, A.** Origins of Parameters in Adimensional Models. *Int. J. Theor. Phys.* [80%; Q1; CS: 2.5] **62**, 198, DOI: [10.1007/s10773-023-05456-z](https://doi.org/10.1007/s10773-023-05456-z) (2023). [arXiv:2302.04076], **1 cite**
- 2022 **Q1** [19] Athron, P., Balázs, C., **FOWLIE, A.**, Morris, L., White, G. & Zhang, Y. How arbitrary are perturbative calculations of the electroweak phase transition? *JHEP* [94%; Q1; CS: 10] **01**, 050, DOI: [10.1007/JHEP01\(2023\)050](https://doi.org/10.1007/JHEP01(2023)050) (2023). [arXiv:2208.01319], **60 cites TOPCITE50+**
- ★ **Q1** [20] Ashton, G. *et al.* Nested sampling for physical scientists. *Nature Reviews Methods Primers* [98%; Q1; CS: 46.1] **2**, DOI: [10.1038/s43586-022-00121-x](https://doi.org/10.1038/s43586-022-00121-x) (2022). [arXiv:2205.15570], **818 cites TOPCITE100+**
- ★ **Q1** [21] Athron, P., **FOWLIE, A.**, Lu, C.-T., Wu, L., Wu, Y. & Zhu, B. Hadronic uncertainties versus new physics for the W boson mass and Muon $g - 2$ anomalies. *Nature Commun.* [97%; Q1; CS: 24.9] **14**, 659, DOI: [10.1038/s41467-023-36366-7](https://doi.org/10.1038/s41467-023-36366-7) (2023). [arXiv:2204.03996], **113 cites TOPCITE100+**
- Q1** [22] Athron, P., Balázs, C., **FOWLIE, A.**, Lv, H., Su, W., Wu, L., Yang, J. M. & Zhang, Y. Global fits of SUSY at future Higgs factories. *Phys. Rev. D* [91%; Q1; CS: 8.3] **105**, 115029, DOI: [10.1103/PhysRevD.105.115029](https://doi.org/10.1103/PhysRevD.105.115029) (2022). [arXiv:2203.04828], **11 cites**
- 2021 ★ **Q2** [23] **FOWLIE, A.** Neyman-Pearson lemma for Bayes factors. *Commun. In Stat. — Theory Method* [51%; Q2; CS: 2.0] 1–8, DOI: [10.1080/03610926.2021.2007265](https://doi.org/10.1080/03610926.2021.2007265) (2021). [arXiv:2110.15625], **7 cites**
- ★ **Q1** [24] **FOWLIE, A.** Comment on “Accumulating evidence for the associate production of a neutral scalar with mass around 151 GeV”. *Phys. Lett. B* [92%; Q1; CS: 9.1] **827**, 136936, DOI: [10.1016/j.physletb.2022.136936](https://doi.org/10.1016/j.physletb.2022.136936) (2022). [arXiv:2109.13426], **8 cites**
- Q1** [25] Cranmer, K. *et al.* Publishing statistical models: Getting the most out of particle physics experiments. *SciPost Phys.* [87%; Q1; CS: 8.2] **12**, 037, DOI: [10.21468/SciPostPhys.12.1.037](https://doi.org/10.21468/SciPostPhys.12.1.037) (2022). [arXiv:2109.04981], **56 cites**
- Q1** [26] Athron, P. *et al.* Thermal WIMPs and the scale of new physics: global fits of Dirac dark matter effective field theories. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] **81**, 992, DOI: [10.1140/epjc/s10052-021-09712-6](https://doi.org/10.1140/epjc/s10052-021-09712-6) (2021). [arXiv:2106.02056], **52 cites TOPCITE50+**

- ★ **Q1** [27] **FOWLIE, A.**, Hoof, S. & Handley, W. Nested Sampling for Frequentist Computation: Fast Estimation of Small p-Values. *Phys. Rev. Lett.* [94%; **Q1**; CS: 16.5] **128**, 021801, DOI: [10.1103/PhysRevLett.128.021801](https://doi.org/10.1103/PhysRevLett.128.021801) (2022). [[arXiv:2105.13923](https://arxiv.org/abs/2105.13923)], **7 cites**
- ★ [28] **FOWLIE, A.** Comment on “Reproducibility and Replication of Experimental Particle Physics Results”. *Harvard Data Science Review* DOI: [10.1162/99608f92.b9bfc518](https://doi.org/10.1162/99608f92.b9bfc518) (2021). [[arXiv:2105.03082](https://arxiv.org/abs/2105.03082)], **2 cites**
- Q1** [29] Balázs, C. *et al.* A comparison of optimisation algorithms for high-dimensional particle and astrophysics applications. *JHEP* [94%; **Q1**; CS: 10] **05**, 108, DOI: [10.1007/JHEP05\(2021\)108](https://doi.org/10.1007/JHEP05(2021)108) (2021). [[arXiv:2101.04525](https://arxiv.org/abs/2101.04525)], **33 cites**
- 2020 ★ **Q2** [30] **FOWLIE, A.** Objective Bayesian approach to the Jeffreys-Lindley paradox. *Commun. In Stat. — Theory Method* [51%; **Q2**; CS: 2.0] 1–6, DOI: [10.1080/03610926.2020.1866206](https://doi.org/10.1080/03610926.2020.1866206) (2020). [[arXiv:2012.04879](https://arxiv.org/abs/2012.04879)], **81 cite**
- ★ **Q1** [31] AbdusSalam, S. S. *et al.* Simple and statistically sound recommendations for analysing physical theories. *Rept. Prog. Phys.* [98%; **Q1**; CS: 31.9] **85**, 052201, DOI: [10.1088/1361-6633/ac60ac](https://doi.org/10.1088/1361-6633/ac60ac) (2022). [[arXiv:2012.09874](https://arxiv.org/abs/2012.09874)], **35 cites**
- ★ **Q1** [32] **FOWLIE, A.**, Handley, W. & Su, L. Nested sampling with plateaus. *Mon. Not. Roy. Astron. Soc.* [87%; **Q1**; CS: 9.1] **503**, 1199–1205, DOI: [10.1093/mnras/stab590](https://doi.org/10.1093/mnras/stab590) (2021). [[arXiv:2010.13884](https://arxiv.org/abs/2010.13884)], **11 cites**
- ★ **Q1** [33] Athron, P. *et al.* Global fits of axion-like particles to XENON1T and astrophysical data. *JHEP* [94%; **Q1**; CS: 10] **05**, 159, DOI: [10.1007/JHEP05\(2021\)159](https://doi.org/10.1007/JHEP05(2021)159) (2021). [[arXiv:2007.05517](https://arxiv.org/abs/2007.05517)], **66 cites** **TOPCITE50+**
- ★ **Q1** [34] **FOWLIE, A.**, Handley, W. & Su, L. Nested sampling cross-checks using order statistics. *Mon. Not. Roy. Astron. Soc.* [87%; **Q1**; CS: 9.1] **497**, 5256–5263, DOI: [10.1093/mnras/staa2345](https://doi.org/10.1093/mnras/staa2345) (2020). [[arXiv:2006.03371](https://arxiv.org/abs/2006.03371)], **22 cites**
- ★ **Q1** [35] Athron, P., Balázs, C., **FOWLIE, A.** & Zhang, Y. PhaseTracer: tracing cosmological phases and calculating transition properties. *Eur. Phys. J. C* [91%; **Q1**; CS: 8.1] **80**, 567, DOI: [10.1140/epjc/s10052-020-8035-2](https://doi.org/10.1140/epjc/s10052-020-8035-2) (2020). [[arXiv:2003.02859](https://arxiv.org/abs/2003.02859)], **58 cites**
- 2019 **Q1** [36] Athron, P., Balázs, C., **FOWLIE, A.**, Pozzo, G., White, G. & Zhang, Y. Strong first-order phase transitions in the NMSSM — a comprehensive survey. *JHEP* [94%; **Q1**; CS: 10] **11**, 151, DOI: [10.1007/JHEP11\(2019\)151](https://doi.org/10.1007/JHEP11(2019)151) (2019). [[arXiv:1908.11847](https://arxiv.org/abs/1908.11847)], **54 cites** **TOPCITE50+**
- ★ **Q2** [37] **FOWLIE, A.** Bayesian and frequentist approaches to resonance searches. *JINST* [59%; **Q2**; CS: 2.4] **14**, P10031, DOI: [10.1088/1748-0221/14/10/P10031](https://doi.org/10.1088/1748-0221/14/10/P10031) (2019). [[arXiv:1902.03243](https://arxiv.org/abs/1902.03243)], **9 cites**
- Q1** [38] Athron, P., Balázs, C., Bardsley, M., **FOWLIE, A.**, Harries, D. & White, G. BubbleProfiler: finding the field profile and action for cosmological phase transitions. *Comput. Phys. Commun.* [93%; **Q1**; CS: 12.1] **244**, 448–468, DOI: [10.1016/j.cpc.2019.05.017](https://doi.org/10.1016/j.cpc.2019.05.017) (2019). [[arXiv:1901.03714](https://arxiv.org/abs/1901.03714)], **82 cites** **TOPCITE50+**
- 2018 ★ **Q1** [39] **FOWLIE, A.** Non-parametric uncertainties in the dark matter velocity distribution. *JCAP* [88%; **Q1**; CS: 10.2] **01**, 006, DOI: [10.1088/1475-7516/2019/01/006](https://doi.org/10.1088/1475-7516/2019/01/006) (2019). [[arXiv:1809.02323](https://arxiv.org/abs/1809.02323)], **12 cites**
- Q1** [40] Athron, P. *et al.* Combined collider constraints on neutralinos and charginos. *Eur. Phys. J. C* [91%; **Q1**; CS: 8.1] **79**, 395, DOI: [10.1140/epjc/s10052-019-6837-x](https://doi.org/10.1140/epjc/s10052-019-6837-x) (2019). [[arXiv:1809.02097](https://arxiv.org/abs/1809.02097)], **114 cites** **TOPCITE100+**
- Q1** [41] Athron, P. *et al.* Global analyses of Higgs portal singlet dark matter models using GAMBIT. *Eur. Phys. J. C* [91%; **Q1**; CS: 8.1] **79**, 38, DOI: [10.1140/epjc/s10052-018-6513-6](https://doi.org/10.1140/epjc/s10052-018-6513-6) (2019). [[arXiv:1808.10465](https://arxiv.org/abs/1808.10465)], **153 cites** **TOPCITE100+**
- ★ **Q1** [42] **FOWLIE, A.** A fast C++ implementation of thermal functions. *Comput. Phys. Commun.* [93%; **Q1**; CS:

- 12.1] **228**, 264–272, DOI: [10.1016/j.cpc.2018.02.015](https://doi.org/10.1016/j.cpc.2018.02.015) (2018). [[arXiv:1802.02720](https://arxiv.org/abs/1802.02720)], **21 cites**
- 2017 ★ **Q1** [43] **FOWLIE, A.** DAMPE squib? Significance of the 1.4 TeV DAMPE excess. *Phys. Lett. B* [92%; **Q1**; **CS: 9.1**] **780**, 181–184, DOI: [10.1016/j.physletb.2018.03.006](https://doi.org/10.1016/j.physletb.2018.03.006) (2018). [[arXiv:1712.05089](https://arxiv.org/abs/1712.05089)], **34 cites**
- Q1** [44] Athron, P., Balázs, C., **FOWLIE, A.** & Zhang, Y. Model-independent analysis of the DAMPE excess. *JHEP* [94%; **Q1**; **CS: 10**] **02**, 121, DOI: [10.1007/JHEP02\(2018\)121](https://doi.org/10.1007/JHEP02(2018)121) (2018). [[arXiv:1711.11376](https://arxiv.org/abs/1711.11376)], **57 cites TOPCITE50+**
- ★ **Q1** [45] Ellis, J., **FOWLIE, A.**, Marzola, L. & Raidal, M. Statistical Analyses of Higgs- and Z-Portal Dark Matter Models. *Phys. Rev. D* [91%; **Q1**; **CS: 8.3**] **97**, 115014, DOI: [10.1103/PhysRevD.97.115014](https://doi.org/10.1103/PhysRevD.97.115014) (2018). [[arXiv:1711.09912](https://arxiv.org/abs/1711.09912)], **51 cites**
- Q1** [46] Athron, P., Balázs, C., Farmer, B., **FOWLIE, A.**, Harries, D. & Kim, D. Bayesian analysis and naturalness of (Next-to-)Minimal Supersymmetric Models. *JHEP* [94%; **Q1**; **CS: 10**] **10**, 160, DOI: [10.1007/JHEP10\(2017\)160](https://doi.org/10.1007/JHEP10(2017)160) (2017). [[arXiv:1709.07895](https://arxiv.org/abs/1709.07895)], **23 cites**
- ★ **Q1** [47] **FOWLIE, A.** Halo-independence with quantified maximum entropy at DAMA/LIBRA. *JCAP* [88%; **Q1**; **CS: 10.2**] **10**, 002, DOI: [10.1088/1475-7516/2017/10/002](https://doi.org/10.1088/1475-7516/2017/10/002) (2017). [[arXiv:1708.00181](https://arxiv.org/abs/1708.00181)], **13 cites**
- Q1** [48] Di Chiara, S., **FOWLIE, A.**, Fraser, S., Marzo, C., Marzola, L., Raidal, M. & Spethmann, C. Minimal flavor-changing Z' models and muon $g - 2$ after the R_{K^*} measurement. *Nucl. Phys. B* [76%; **Q1**; **CS: 5.5**] **923**, 245–257, DOI: [10.1016/j.nuclphysb.2017.08.003](https://doi.org/10.1016/j.nuclphysb.2017.08.003) (2017). [[arXiv:1704.06200](https://arxiv.org/abs/1704.06200)], **71 cites TOPCITE50+**
- 2016 **Q1** [49] Balázs, C., **FOWLIE, A.**, Mazumdar, A. & White, G. Gravitational waves at aLIGO and vacuum stability with a scalar singlet extension of the Standard Model. *Phys. Rev. D* [91%; **Q1**; **CS: 8.3**] **95**, 043505, DOI: [10.1103/PhysRevD.95.043505](https://doi.org/10.1103/PhysRevD.95.043505) (2017). [[arXiv:1611.01617](https://arxiv.org/abs/1611.01617)], **61 cites TOPCITE50+**
- ★ **Q1** [50] **FOWLIE, A.** Bayes factor of the ATLAS diphoton excess: Using Bayes factors to understand anomalies at the LHC. *Eur. Phys. J. Plus* [75%; **Q1**; **CS: 5.4**] **132**, 46, DOI: [10.1140/epjp/i2017-11340-1](https://doi.org/10.1140/epjp/i2017-11340-1) (2017). [[arXiv:1607.06608](https://arxiv.org/abs/1607.06608)], **5 cites**
- Q2** [51] Bianchini, L., Calpas, B., Conway, J., **FOWLIE, A.**, Marzola, L., Veelken, C. & Perrini, L. Reconstruction of the Higgs mass in events with Higgs bosons decaying into a pair of τ leptons using matrix element techniques. *Nucl. Instrum. Meth. A* [56%; **Q2**; **CS: 3.2**] **862**, 54–84, DOI: [10.1016/j.nima.2017.05.001](https://doi.org/10.1016/j.nima.2017.05.001) (2017). [[arXiv:1603.05910](https://arxiv.org/abs/1603.05910)], **52 cites**
- ★ **Q1** [52] **FOWLIE, A.** & Bardsley, M. H. Superplot: a graphical interface for plotting and analysing MultiNest output. *Eur. Phys. J. Plus* [75%; **Q1**; **CS: 5.4**] **131**, 391, DOI: [10.1140/epjp/i2016-16391-0](https://doi.org/10.1140/epjp/i2016-16391-0) (2016). [[arXiv:1603.00555](https://arxiv.org/abs/1603.00555)], **38 cites**
- ★ **Q1** [53] **FOWLIE, A.**, Balázs, C., White, G., Marzola, L. & Raidal, M. Naturalness of the relaxion mechanism. *JHEP* [94%; **Q1**; **CS: 10**] **08**, 100, DOI: [10.1007/JHEP08\(2016\)100](https://doi.org/10.1007/JHEP08(2016)100) (2016). [[arXiv:1602.03889](https://arxiv.org/abs/1602.03889)], **30 cites**
- 2015 ★ **Q1** [54] **FOWLIE, A.** & Marzola, L. Examining a right-handed quark mixing matrix with b -tags at the LHC. *Nucl. Phys. B* [76%; **Q1**; **CS: 5.5**] **894**, 588–601, DOI: [10.1016/j.nuclphysb.2015.03.025](https://doi.org/10.1016/j.nuclphysb.2015.03.025) (2015). [[arXiv:1412.5587](https://arxiv.org/abs/1412.5587)], **4 cites**
- 2014 ★ **Q1** [55] **FOWLIE, A.** & Marzola, L. Testing quark mixing in minimal left–right symmetric models with b -tags at the LHC. *Nucl. Phys. B* [76%; **Q1**; **CS: 5.5**] **889**, 36–45, DOI: [10.1016/j.nuclphysb.2014.10.009](https://doi.org/10.1016/j.nuclphysb.2014.10.009) (2014). [[arXiv:1408.6699](https://arxiv.org/abs/1408.6699)], **15 cites**
- ★ **Q1** [56] **FOWLIE, A.** Is the CNMSSM more credible than the CMSSM? *Eur. Phys. J. C* [91%; **Q1**; **CS: 8.1**] **74**, 3105, DOI: [10.1140/epjc/s10052-014-3105-y](https://doi.org/10.1140/epjc/s10052-014-3105-y) (2014). [[arXiv:1407.7534](https://arxiv.org/abs/1407.7534)], **23 cites**
- ★ **Q1** [57] **FOWLIE, A.** CMSSM, naturalness and the “fine-tuning price” of the Very Large Hadron Collider. *Phys.*

Rev. D [91%; Q1; CS: 8.3] **90**, 015010, DOI: [10.1103/PhysRevD.90.015010](https://doi.org/10.1103/PhysRevD.90.015010) (2014). [[arXiv:1403.3407](https://arxiv.org/abs/1403.3407)], **45 cites**

- ★ Q1 [58] **FOWLIE, A.** & Raidal, M. Prospects for constrained supersymmetry at $\sqrt{s} = 33$ TeV and $\sqrt{s} = 100$ TeV proton-proton super-colliders. *Eur. Phys. J. C* [91%; Q1; CS: 8.1] **74**, 2948, DOI: [10.1140/epjc/s10052-014-2948-6](https://doi.org/10.1140/epjc/s10052-014-2948-6) (2014). [[arXiv:1402.5419](https://arxiv.org/abs/1402.5419)], **34 cites**

2013 Q1 [59] **FOWLIE, A.**, Kowalska, K., Roszkowski, L., Sessolo, E. M. & Tsai, Y.-L. S. Dark matter and collider signatures of the MSSM. *Phys. Rev. D* [91%; Q1; CS: 8.3] **88**, 055012, DOI: [10.1103/PhysRevD.88.055012](https://doi.org/10.1103/PhysRevD.88.055012) (2013). [[arXiv:1306.1567](https://arxiv.org/abs/1306.1567)], **94 cites TOPCITE50+**

2012 Q1 [60] **FOWLIE, A.**, Kazana, M., Kowalska, K., Munir, S., Roszkowski, L., Sessolo, E. M., Trojanowski, S. & Tsai, Y.-L. S. The CMSSM Favoring New Territories: The Impact of New LHC Limits and a 125 GeV Higgs. *Phys. Rev. D* [91%; Q1; CS: 8.3] **86**, 075010, DOI: [10.1103/PhysRevD.86.075010](https://doi.org/10.1103/PhysRevD.86.075010) (2012). [[arXiv:1206.0264](https://arxiv.org/abs/1206.0264)], **186 cites TOPCITE100+**

2011 Q1 [61] **FOWLIE, A.**, Kalinowski, A., Kazana, M., Roszkowski, L. & Tsai, Y. L. S. Bayesian Implications of Current LHC and XENON100 Search Limits for the Constrained MSSM. *Phys. Rev. D* [91%; Q1; CS: 8.3] **85**, 075012, DOI: [10.1103/PhysRevD.85.075012](https://doi.org/10.1103/PhysRevD.85.075012) (2012). [[arXiv:1111.6098](https://arxiv.org/abs/1111.6098)], **64 cites TOPCITE50+**

RESEARCH SOFTWARE

- **Scientific computing in particle physics and cosmology:** [BubbleProfiler](#), [PhaseTracer](#), [TransitionSolver](#) and [thermal_funcs](#)
- **Global analyses of high-energy physics models:** [GAMBIT](#)
- **Statistical computing:** [stanhf](#) (Stan implementation of HistFactory models), [polystan](#) (nested sampling for Stan models), and [kingpin](#) (treed Gaussian processes)
- **Experimental/laboratory software:** [cmdpasco](#) (data acquisition from PASCO smart carts)
- **Research productivity tools:** [bibiso4](#), [texcheck](#), and [bibtexify](#)
- **Example of open-source scientific software contributions:** [SciPy](#) (merged pull request of efficient computation of moments of normal distribution)

TALKS & SEMINARS

- Over 50 presentations and talks
- Slides: <https://andrewfowlie.github.io/talk/>

Invited

- 2026 [1] [Mapping the ALPs: A Naturalness Map for Axions and ALPs](#), 2026 Workshop on New Physics and Interdisciplinary Sciences (NPhIS 2026) , 18 May
- 2025 [2] [Are axion solutions to the CP problem fine-tuned?](#), The Fourth International Conference on Axion Physics and Experiment, 29 July
- [3] [Fitting—statistical modeling and analysis for physics](#), Lecture, Third New Physical Numerical Numerical Computing and Simulation Frontier Workshop, Xinxiang, 3 July
- 2024 [4] [A cry of distress from Nature? Fine-tuning in scientific theories](#), XJTLU CHIPS Wisdom Forum, 9 October
- [5] [Cosmological phase transitions: From perturbative particle physics to gravitational waves](#), XJTLU SMP

Research Excellence Workshop, 29 May

[6] [Testing fundamental theories with global fits](#), Seminar, Duke Kunshan, 10 May

[7] [Testing fundamental theories with global fits](#), Seminar, Suzhou University, 26 April

[8] [The Bayes factor surface for searches for new physics](#), NANOGrav New Physics Working Group, 20 February

2023 [9] [Origins of parameters in adimensional models](#), Seminar, Fudan University, 20 October

[10] [From first order phase transitions to gravitational waves](#), The 2023 Shanghai Symposium on Particle Physics and Cosmology, Tsung-Dao Lee Institute, 23 September

[11] [New physics in the garden of forking paths](#), Mini-Workshop on Anomalies at the LHC, Tsung-Dao Lee Institute, 21 September

[12] [Opening up Nested Sampling](#), MaxEnt 2023, 6 July

[13] [Origins of parameters in adimensional models](#), Seminar, Zhejiang University, 2 June

[14] [Origins of parameters in adimensional models](#), Seminar, Shandong University, 24 May

2022 [15] [Herding cats? — Bayesian and frequentist methods and compromises](#), University of Goettingen CATs seminar, 14 May

2021 [16] [Nested sampling for frequentist computation: fast estimation of small \$p\$ -values](#), ATLAS statistics forum, 29 July

[17] [Nested sampling for frequentist computation: fast estimation of small \$p\$ -values](#), Purple Mountain Observatory, 9 July

[18] [Evidence for axion-like particles from XENON1T and astrophysical data](#), NCBJ, Warsaw, 12 January

2020 [19] [Nested sampling cross-checks using order statistics](#), Monash University, 21 July

[20] [Nested sampling cross-checks using order statistics](#), Cambridge University, 15 July

2019 [21] [Strong first-order phase transitions in the NMSSM and methods for finding them](#), SJTU-U. Sydney Workshop on the Electroweak Phase Transition, Tsung-Dao Lee Institute, 19 December

[22] [Bayesian and frequentist approaches to discoveries](#), PASCOS, July 2

[23] [Bayesian and frequentist approaches to resonance searches](#), Purple Mountain Observatory, 16 April

2018 [24] [Statistical Analyses of Higgs- and Z-Portal Dark Matter Models](#), Nanjing Normal University, 25 June

[25] [Statistical Analyses of Higgs- and Z-Portal Dark Matter Models](#), Melbourne University, 8 March

2017 [26] [Relative plausibility of scientific theories: WIMP dark matter](#), Fundamental Physics, Symmetry and Life, University of Sydney, 30 November

[27] [Halo-independence with quantified maximum entropy](#), NCTS Workshop on Dark Matter, Particles and Cosmos, Taiwan, 14 October

[28] [Halo-independence with quantified maximum entropy](#), NTU, Taiwan, 12 October

[29] [Halo-independence with quantified maximum entropy](#), IPMU, Tokyo, 4 October

Other talks

2026 [30] [Mapping the ALPs: A Naturalness Map for Axions and ALPs](#), The Fifth International Conference on Axion Physics and Experiment (Axion 2026), 10 August

- [31] [Mapping the ALPs: A Naturalness Map for Axions and ALPs](#), XJTLU, 10 June
- 2024 [32] [The status of fine-tuning arguments in the CEPC era](#), CEPC New Physics Workshop, 31 August
- 2023 [33] [Origins of parameters in adimensional models](#), Colloquium, XJTLU, School of Maths and Physics, 28 September
- [34] [Origins of parameters in adimensional models](#), MaxEnt 2023, 7 July
- 2021 [35] [Nested sampling for frequentist computation: fast estimation of small \$p\$ -values](#), Computational Tools for High Energy Physics and Cosmology, 26 November
- [36] [Getting the most out of particle physics experiments](#), Workshop on Hadron Structure at High-Energy, High-Luminosity Facilities 2021, 27 October
- [37] [Pitfalls in likelihood land](#), (Re)interpreting the results of new physics searches at the LHC, 18 February
- 2020 [38] [Nested sampling cross-checks using order statistics](#), First International Symposium on the Interdisciplinary Frontiers of Gravity, Matter and Quantum Information, 28 December
- 2019 [39] [Strong first-order phase transitions in the NMSSM and methods for finding them](#), SJTU-U. Sydney Workshop on the Electroweak Phase Transition, 19 December
- [40] [Combined collider constraints on neutralinos and charginos](#), The tenth Weihai New Physics Workshop, Shandong University, 14 August
- [41] [Bayesian and frequentist approaches to resonance searches](#), Fourteenth workshop on TeV physics, Nanjing, 21 April
- [42] [Bayesian and frequentist approaches to resonance searches](#), Nanjing Normal University, 17 April
- 2018 [43] [Non-parametric uncertainties in the dark matter velocity distribution](#), Auckland University, 10 December
- [44] [Statistical Analyses of Higgs- and Z-Portal Dark Matter Models](#), Seoul, ICHEP 2018, 6 July
- [45] [Potential applications of machine learning in particle physics](#), Machine Learning Symposium, National Centre for Synchrotron Science, 19 March
- 2017 [46] [Using Bayes factors to understand anomalies at the LHC](#), Energy Frontier in Particle Physics: LHC and Future Colliders, NTU, Taiwan, 30 September
- 2016 [47] [The Jeffreys-Lindley's Paradox](#), CompStats Meeting, Monash University, 1 November
- [48] [Bayesian approach to naturalness](#), Fine-tuning, the Multiverse and Life, Sydney, 24 November
- [49] [Naturalness of the relaxion mechanism](#), CosPA, Sydney, 29 November
- [50] [Bayesian naturalness of Next-to-Minimal and Minimal Supersymmetric Models](#), SUSY 2016, Melbourne, 5 July
- [51] [Naturalness of the relaxion mechanism](#), SUSY 2016, Melbourne, 7 July
- [52] [Naturalness of the relaxion mechanism](#), CoEPP Annual Theory Meeting, Melbourne, 16 February
- 2014 [53] [Prospects for constrained supersymmetry at \$\sqrt{s} = 33\$ TeV and \$\sqrt{s} = 100\$ TeV proton-proton supercolliders](#), Deep Inelastic Scattering, Warsaw, 29 April
- 2013 [54] [Bayesian reconstruction of SUSY parameters via the golden decay](#), Theory Meets Experiment, Warsaw, 6 October
- [55] [Status of CMSSM after LHC Run-I](#), HEP IOP, Liverpool, 9 April
- 2012 [56] [The CMSSM after 2 years of the LHC](#), Consortium for Fundamental Physics, Sheffield, 9 April

- 2011 [57] [Bayesian Implications of Current LHC Limits for the Constrained MSSM](#), Young Theorists' Forum, Durham, 13 December
- [58] [Supersymmetry and the LHC](#), Seminar, University of Sheffield, 1 October

Teaching & Supervision

PROFESSIONAL DEVELOPMENT

- 2025 ■ [Instructional Skills Workshop](#), XJTLU, **Certificate of Completion**, 24 hours

CURRICULUM DEVELOPMENT

- AY23–24 ■ Representative, Department of Physics, *School Curriculum Review Panel (SCR)*

SUPERVISION AND MENTORSHIP

Postgraduate

- 2025–
AY26–27 ■ Primary supervisor for physics Ph.D. student, Hao Yang
- AY25–26 ■ Supervising MSc Data Science student, Sunxiao Shen
- AY25–26 ■ Supervised MSc Data Science student, Qiao Wen, on noisy optimization methods
- 2022–2024 ■ Supervised [Qiao Li](#) for three-year Master's project on measuring contributions to precision observables using Gaussian processes [[2306.17385](#)]
- 2020–2021 ■ Informal supervisor [Liangliang Su](#) Master's project, writing two papers [[2006.03371](#), [2010.13884](#)]. Now postdoctoral researcher at KIT, Karlsruhe and author of Migdal effect discovery [[Nature 649 \(2026\) 8097, 580-583](#)]
- 2016–2018 ■ Supervised (10%) Ph.D. student, [Giancarlo Pozzo](#), on baryogenesis in next-to-minimal supersymmetric models. My role included QFT tutorials

Undergraduate

- 2023–2026 ■ Supervised 4 SURF summer projects involving over 15 students, including building a Lorenz wheel and a Galton board
- 2023–2026 ■ Supervised 8 Final Year Projects—[MTH301](#). Former student [Qihong Jiang](#) now Ph.D. in stochastic modelling at Queen Mary University of London
- 2022–2023 ■ Supervised 10 undergraduates leading to [[2211.03258](#)]. [Huifang Lyu](#) now Ph.D. at GRAPPA, University of Amsterdam
- 2017–2018 ■ Supervised undergraduate project about the bounce equation and its connection to phase transitions and baryogenesis
- 2015–2016 ■ Supervised undergraduate [Michael Bardsley](#)'s summer project. We developed statistical software resulting in a publication [[1603.00555](#)]. Michael subsequently completed a Ph.D. in particle physics at Monash University (2024)

TEACHING

- AY25–26 ■ Led [PHY002-2526-S5](#) Mechanics — 2.5 credits — 2,701 students. Responsible for overseeing 9 co-teachers, laboratory sessions and assessments

- AY24 – 25
 - Taught [MTH101-2526-S1](#) Engineering Mathematics I — 5 credits — 70 students
 - Taught [PHY001-2425-S2](#) Classical Physics for Engineers — 2.5 credits — 115 students
 - Taught [MTH101-2425-S1](#) Engineering Mathematics I — 5 credits — 196 students
- AY23 – 24
 - Taught [PHY002-2324-S2](#) Physics — 5 credits — 217 students
 - Taught [MTH101-2324-S1](#) Engineering Mathematics I — 5 credits — 140 students
- AY22 – 23
 - Taught [PHY002-2223-S2](#) Physics — 5 credits — 123 students
- 2019 – 2022
 - Post-graduate course on physics beyond the Standard Model — about 20 students and about 25 hours
- 2015
 - Lectures on statistics for physicists at the University of Tartu — six hours and about 5 students
- 2012 – 2013
 - First-year physics tutor, weekly tutorials — about 20 sessions with about 10 students
- 2010 – 2012
 - Undergraduate physics weekly problem class assistant — about 30 students

Service

MEDIA

Articles in international news outlets, blogs and our University Marketing and Communications (UMC). Coauthored [article](#) about [Phys. Rev. Lett.](#) that was syndicated in 212 non-Chinese media outlets in 7 languages with a total of about 700 000 unique readers per month. The media exposure for this alone corresponds to an advertising value equivalency of about **\$1.4 million USD**. For more details see the [UMC 30 day media report](#).

Selected internal news articles

- [Dr. Andrew Fowlie joins NANOGrav](#)
- [Dr. Andrew Fowlie publishes a high-level review](#)
- [Nanohertz gravitational waves are cool but not supercool](#)
- [SMP wins 8 NSFC grants in 2024](#)
- [28 projects approved](#)

Selected external news articles

- [Phys.org](#) and [Monash University press](#) about [Prog. Part. Nucl. Phys.](#)
- [Phys.org](#) about [Phys. Rev. Lett.](#) paper
- [Big Think](#) about [Nature Commun.](#) paper
- [ABC News \(Norway\)](#) about [Eur. Phys. J. C](#) paper

DEPARTMENTAL

- 2024 –
 - *Departmental Events Officer*. Organizing regular seminars – see [Seminars @ Department of Physics](#). Growing connections to other academic units. More than 30 seminars in last three years
- 2023 –
 - *Departmental IT Officer*. Initiated and led department webpage update to raise departmental profile — see [Department of Physics](#)
- 2019 – 2023
 - Built and maintained group [webpage](#)
 - Organized [online seminar series](#), including [event with Nobel Laureate](#) with audience of over 3,500

COLLABORATIONS

- 2023 – CIRCULAR ELECTRON-POSITRON COLLIDER (CEPC) — A proposed next-generation world-leading experiment in fundamental science [GAMBIT](#) — International collaboration performing statistical analyses of models of new physics
- 2016 –
- 2024 – 2025 [NANOGrav](#) — World-leading pulsar-timing array search for gravitational waves to contribute towards data analysis
- 2011 – 2013 [BAYESFITS](#) — Bayesian analyses of supersymmetric models in light of first run of LHC, led by [Prof. Roszkowski](#)

EDITORIAL

Referee for **physics journals**: *Nature Commun.*, *Phys. Rev. Lett.*, *Phys. Rev. D*, *Eur. Phys. J. C*, *J. Phys. G*, *Ann. Phys.*, *Metrologia*, *Nucl. Phys. B*, [SciPost Physics](#), and *Int. J. Mod. Phys. A*

Referee for **statistics journals**: *Stat. Pap.*, and *Stat. & Comput.*

Referee for **software journals**: [J. Open Source Softw.](#) and *Comput. Phys. Comm.*

- 2020 – 2023 Editor for [Journal of Nanjing Normal University, Physical Sciences](#)

UNIVERSITY

- 2025 ■ Internal peer reviewer for 2025 NSFC RFIS candidates, June 9
- 2024 ■ Presented at Experience Sharing for Research Fund for International Scientists, 11th December
- 2023 ■ Jiangsu CEAA Interviewer at XJTLU — 2 days
- 2023 – 2025 ■ Coordinated University staff social football & tennis clubs

Education and other relevant experience

- 2009 – 2013 Ph.D., University of Sheffield, UK
[Bayesian Approach to Investigating Supersymmetric Models](#). Supervised by [Prof. Roszkowski](#). Viva passed with minor corrections, examined by [Prof. King \(University of Southampton\)](#) and [Prof. van de Bruck \(University of Sheffield\)](#)
- 2009 – 2010 [Scuola Internazionale Superiore di Studi Avanzati \(SISSA\)](#), Trieste, Italy
Six-month placement studying advanced topics in particle physics and related subjects
- 2005 – 2009 M. PHYS, University of Durham, UK
First-class four-year undergraduate Master's in Physics. Final-year modules included Advanced Theoretical Physics (82%) and Particle Theory (90%). Master's project, *The Search for Dark Matter at the Linear Collider*, supervised by [Prof. Moortgat-Pick](#) (73%)
- 2006 & 2007 Summer placement at electricity supplier E.ON about numerical simulation of atmosphere with parallel computing